

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA43

COURSE OUTCOME COVERED

CO-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. **(L2)**

Assessment Tool: Code along exercise: Make your first HTML page together with CSS

Weightage: 10%

QUESTIONS: Total Marks: 10

Scenario: Web Development Standards Implementation

A start up is developing a modern web application. During development, the team encounters multiple issues related to HTML elements, HTTP methods, XML syntax, and web communication protocols.

The following observations are made:

- The team needs to embed a promotional video on the homepage.
- A registration form must securely send user data to the server.
- The navigation menu must follow semantic HTML standards.
- XML configuration files are being used but contain syntax errors.
- The application must communicate over the correct web protocol.

Data Snippets Provided

Snippet 1: Video Embedding Options

A. `<media src="video.mp4"></media>`

- B. `<video src="video.mp4" controls></video>`
- C. `<movie src="video.mp4"></movie>`
- D. `<vid src="video.mp4"></vid>`

Snippet 2: Form Submission Methods

- A. GET
- B. POST
- C. PUT
- D. DELETE

Snippet 3: Navigation Elements

- A. `<div>`
- B. `<nav>`
- C. `<section>`
- D. `<header>`

Snippet 4: XML Syntax

- A. `<note><to>User</to><from>Admin</from>`
- B. `<note><to>User</to><from>Admin</from></note>`
- C. `<note><to>User</from></note>`
- D. `<note to="User"></note>`

Snippet 5: Web Communication Protocols

- A. FTP
- B. HTTP
- C. SMTP
- D. SNMP

Questions:

1. Select the correct HTML tag for video embedding and justify your answer.

Answer: Option B (`<video src="video.mp4" controls></video>`)

Justification: The HTML5 standard introduced the native `<video>` element to support video playback directly within the browser without requiring external plugins. The 'controls' attribute is essential as it renders standard media controls (play, pause, volume, etc.). Other options like `<media>`, `<movie>`, or `<vid>` are non-standard and invalid tags.

2. Identify the appropriate HTTP method for form submission and explain why.

Answer: Option B (POST)

Justification: The POST method sends the form parameters within the body of the HTTP request payload, keeping them hidden from the URL query string. This is crucial for securely transmitting user registration details, as GET exposes the data in plain text in browser histories, bookmarks, and web server logs.

3. Choose the proper semantic element for navigation and justify its usage.

Answer: Option B (`<nav>`)

Justification: The `<nav>` tag is a dedicated HTML5 semantic element designed to wrap a block of navigation links. Using `<nav>` rather than generic wrappers like `<div>` helps search engines index the page correctly and allows assistive technologies (screen readers) to identify the navigation section for improved accessibility.

4. Identify the correct XML syntax and explain the error in other options.

Answer: Option B (<note><to>User</to><from>Admin</from></note>)

Justification:

- Option B is correct because the root element <note> and child elements <to> and <from> are properly closed and nested.
- Errors in other options: Option A lacks a closing </note> tag for the root element. Option C contains a nesting overlap error (<to>User</from>). Option D lacks the closing '>' bracket on the closing tag (</note>).

5. Choose the correct protocol for web communication and justify your choice.

Answer: Option B (HTTP / HTTPS)

Justification: HTTP (Hypertext Transfer Protocol) and its secure extension HTTPS are the standard application layer protocols designed for data transmission on the World Wide Web. Other options serve different services: FTP is for raw file transfers, SMTP is for sending emails, and SNMP is for network device management.

Code of Conduct:

I G. Abhinay (192373062) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

G.Abhinay

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding ; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding ; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand